Appendix B. Simple Connectivity and Jordan Curve Theorem

Equivalent forms of simple connectivity, the Jordan curve theorem, general Cauchy theorem

Contents

Overview

Simple connectivity underlies many basic results of complex analysis. Recall (Chapter 3) that a region Ω is *simply connected* if any two curves in Ω with the same end-points are homotopic; from this we deduced that

$$\int_\eta f(\zeta)\,d\zeta = 0 \tag{1}$$

for every closed curve $\eta \subset \Omega$ and every f holomorphic in Ω. In this appendix we prove:

(I) Ω is simply connected if and only if it is *holomorphically simply connected*, i.e. (1) holds for all closed $\eta \subset \Omega$ and all f holomorphic in Ω.

(II) A bounded region is simply connected if and only if its complement is connected.

(III) Ω is simply connected if and only if no curve in Ω winds around a point of Ω^c.

In the second part we answer: given a simple closed curve γ, what is the region "enclosed by γ"? We prove the Jordan curve theorem for piecewise-smooth curves: γ^c has exactly two components, the bounded one is simply connected and both have boundary γ. This yields a general form of Cauchy's theorem extending the version for toy contours of Chapter 2. As in Chapter 1, "curve" means "piecewise-smooth curve" unless stated otherwise.

1 Equivalent formulations of simple connectivity

Motivation. Cauchy's theorem holds in a region when every closed curve can be shrunk to a point, but in practice one wants other ways to check this: by winding numbers, or by the complement being connected. This section proves that these conditions all agree.

Theorem 1.1

A region Ω is holomorphically simply connected if and only if Ω is simply connected.

Proof. If Ω is simply connected, it is holomorphically simply connected by Corollary 5.3 of Chapter 3. Conversely, let Ω be holomorphically simply connected. If $\Omega = \mathbb{C}$ it is simply connected. Otherwise the proof of the Riemann mapping theorem (Theorem 3.1) applies verbatim: the only use of simple connectivity there is to construct a holomorphic logarithm (hence square root) of a non-vanishing holomorphic function g, namely $\log g(z) = c + \int_{z_0}^{z} g'/g$, which is well defined precisely because (1) holds. Hence Ω is conformally equivalent to $\mathbb{D}$. Since $\mathbb{D}$ is simply connected and a conformal map carries homotopies to homotopies, Ω is simply connected. ∎

Theorem 1.2

If Ω is a bounded region in $\mathbb{C}$, then Ω is simply connected if and only if Ω^c is connected.

Boundedness is needed: an infinite strip is simply connected but its complement has two components. If complements are taken in the Riemann sphere, the conclusion holds for all regions.

Proof that Ω^c connected $\Rightarrow$ Ω simply connected. By Theorem 1.1 it suffices to show $\int_\eta f = 0$ for $\eta \subset \Omega$ closed and f holomorphic in Ω. With $d(z, \Omega^c) = \inf\{|z - w| : w \in \Omega^c\}$, let

$$K_\delta = \{z \in \overline{\Omega} : d(z, \Omega^c) \geq \delta\}.$$

K_δ is compact (Ω is bounded), $K_\delta \subset \Omega$, and $\eta \subset K_\delta$ for small δ since η is compact. We show that K_δ^c is connected.

Suppose $K_\delta^c = O_1 \cup O_2$ with O_1, O_2 disjoint, open, non-empty, and set $F_i = O_i \cap \Omega^c$. Since $\Omega^c \subset K_\delta^c$, $\Omega^c = F_1 \cup F_2$ with $F_1 \cap F_2 = \emptyset$.

Step F_1 is closed. Let $z_n \in F_1$, $z_n \to z$. Then $z \in \Omega^c$ (Ω^c is closed), so $z \in O_1 \cup O_2$. If $z \in O_2$, then $z_n \in O_2$ for large n as O_2 is open, contradicting $z_n \in O_1$. So $z \in O_1$, i.e. $z \in F_1$. Likewise F_2 is closed.

Step $F_1 \neq \emptyset$. Otherwise $\Omega^c \subset O_2$. Pick $w \in O_1$. Since $w \notin K_\delta$, there is $z \in \Omega^c$ with $|w - z| < \delta$. Every point of the segment $[w, z]$ is within distance $< \delta$ of $z \in \Omega^c$, hence lies in K_δ^c. So the connected segment lies in $O_1 \cup O_2$ and meets both ($w \in O_1$, $z \in O_2$), a contradiction. Likewise $F_2 \neq \emptyset$.

Thus Ω^c is disconnected, contrary to hypothesis; so K_δ^c is connected. By Runge's theorem (Theorem 5.7 of Chapter 2) there are polynomials $P_n \to f$ uniformly on K_δ, hence on η. Since $\int_\eta P_n = 0$ (polynomials have primitives), $\int_\eta f = \lim \int_\eta P_n = 0$. ∎

The converse is proved after Theorem 1.4, using winding numbers.

Winding numbers

If γ is closed and $z \notin \gamma$, each loop of γ around z changes $\frac{1}{2\pi} \arg(\zeta - z)$ by ± 1. Since $\log w = \log|w| + i \arg w$, the quantity $\frac{1}{2\pi i} \int_\gamma \frac{d\zeta}{\zeta - z}$ should count these loops.

Definition (Winding number)

The **winding number** (or **index**) of a closed curve γ around $z \notin \gamma$ is

$$W_\gamma(z) = \frac{1}{2\pi i} \int_\gamma \frac{d\zeta}{\zeta - z}.$$

For $\gamma(t) = e^{ikt}$, $0 \le t \le 2\pi$ ($k \in \mathbb{N}$), $W_\gamma(z) = k$ for $|z| < 1$ and 0 for $|z| > 1$; for e^{-ikt} the values are $-k$ and 0. For a positively oriented toy contour, $W_\gamma = 1$ in its interior and 0 in its exterior.

Lemma 1.3

Let γ be a closed curve in $\mathbb{C}$.

(i) If $z \notin \gamma$, then $W_\gamma(z) \in \mathbb{Z}$.

(ii) If z, w lie in the same connected component of γ^c, then $W_\gamma(z) = W_\gamma(w)$.

(iii) If z lies in the unbounded component of γ^c, then $W_\gamma(z) = 0$.

Proof. (i) Let $\gamma : [0, 1] \to \mathbb{C}$ parametrize the curve and $G(t) = \int_0^t \frac{\gamma'(s)}{\gamma(s) - z}\, ds$. G is continuous and, except at finitely many points, differentiable with $G' = \gamma'/(\gamma - z)$. Then $H(t) = (\gamma(t) - z)e^{-G(t)}$ is continuous with $H' = \gamma' e^{-G} - (\gamma - z)G' e^{-G} = 0$ except at finitely many points, so H is constant: $H(1) = H(0) = \gamma(0) - z$. As $\gamma(1) = \gamma(0) \neq z$, $e^{-G(1)} = 1$, so $G(1) = 2\pi i\, W_\gamma(z) \in 2\pi i\mathbb{Z}$.

(ii) W_γ is a continuous function of $z \in \gamma^c$ (the integrand depends continuously on z, uniformly on γ), integer-valued by (i), hence constant on each connected component.

(iii) $|W_\gamma(z)| \le \frac{\ell(\gamma)}{2\pi\, d(z,\gamma)} \to 0$ as $|z| \to \infty$; since W_γ is constant on the unbounded component by (ii), it is 0 there. ∎

Theorem 1.4

A bounded region Ω is simply connected if and only if $W_\gamma(z) = 0$ for every closed curve $\gamma \subset \Omega$ and every $z \notin \Omega$.

Proof. If Ω is simply connected and $z \notin \Omega$, then $f(\zeta) = 1/(\zeta - z)$ is holomorphic in Ω, and Cauchy's theorem gives $W_\gamma(z) = 0$.

Conversely, by the part of Theorem 1.2 already proved, it suffices to show Ω^c is connected. Suppose not: $\Omega^c = F_1 \cup F_2$ with F_1, F_2 disjoint, closed, non-empty. Only one of them can be unbounded (Ω is bounded, so Ω^c contains the connected set $\{|z| > R\}$), so we may assume F_1 is bounded, hence compact. We construct a closed curve $\gamma \subset \Omega$ and $w \in F_1$ with $W_\gamma(w) \neq 0$, using the following lemma. ∎

Lemma 1.5

Under the above assumptions, let $w \in F_1$. There is a finite collection $\mathcal{Q} = \{Q_1, \dots, Q_n\}$ of closed squares of a uniform grid $\mathcal{G}$ such that:

(i) w is in the interior of Q_1;

(ii) the interiors of Q_j and Q_k are disjoint for $j \neq k$;

(iii) F_1 is contained in the interior of $\bigcup_j Q_j$;

(iv) $\bigcup_j Q_j$ is disjoint from F_2;

(v) the boundary of $\bigcup_j Q_j$ lies in Ω and is a finite union of disjoint simple closed polygonal curves.

Proof of Theorem 1.4, assuming Lemma 1.5. Orient each ∂Q_j positively. Since w is interior to Q_1 and $w \notin Q_j$ for $j > 1$,

$$\sum_{j=1}^{n} \frac{1}{2\pi i} \int_{\partial Q_j} \frac{d\zeta}{\zeta - w} = 1. \tag{2}$$

A side shared by two squares of $\mathcal{Q}$ is traversed twice in opposite directions and cancels. The remaining sides, oriented with $\bigcup Q_j$ on the left, form the polygons $\gamma_1, \dots, \gamma_M$ of (v), so $\sum_{j=1}^{M} W_{\gamma_j}(w) = 1$. Hence $W_{\gamma_{j_0}}(w) \neq 0$ for some j_0, while $\gamma_{j_0} \subset \Omega$ and $w \notin \Omega$: a contradiction. ∎

Proof of Lemma 1.5. F_1 is compact and F_2 closed, so $d = d(F_1, F_2) > 0$. Take a grid $\mathcal{G}_0$ of closed squares of side $\delta < d/100$ such that w is the center of a square R_1, and let $\mathcal{R} = \{R_1, \dots, R_m\}$ be the squares of $\mathcal{G}_0$ meeting F_1.

Step (i)–(iv) and $\partial \bigcup R_j \subset \Omega$. (i), (ii) are clear. Every point of $\bigcup R_j$ is within $\sqrt{2}\delta < d$ of F_1, giving (iv). If $z \in F_1$, every square of $\mathcal{G}_0$ containing z meets F_1 and so belongs to $\mathcal{R}$; these squares cover a neighborhood of z, giving (iii). A *boundary side* (a side of exactly one square of $\mathcal{R}$) is a side of a square not in $\mathcal{R}$, so it misses F_1; it also misses F_2 by (iv). Hence $\partial \bigcup R_j \subset \Omega$.

Step Bad vertices. The boundary of $\bigcup R_j$ is the union of the boundary sides; their end-points are boundary vertices. At a grid vertex P the four surrounding squares give 0, 2 or 4 boundary sides at P, and 4 occurs exactly when two diagonally opposite squares are in $\mathcal{R}$ and the other two are not. Call such a P *bad*.

Step Refinement. Let $\mathcal{G}$ be the grid obtained by cutting each square of $\mathcal{G}_0$ into 9 equal subsquares, and take all $9m$ subsquares of squares of $\mathcal{R}$, with Q_1 the middle subsquare of R_1 (so w is its center). At each bad vertex P add the two subsquares of $\mathcal{G}$ having P as a corner and lying in the two missing squares of $\mathcal{G}_0$. Call the result $\mathcal{Q} = \{Q_1, \dots, Q_n\}$. Now all four small squares at P are in $\mathcal{Q}$, so P is interior. An added subsquare s at P has three other corners. At the two lying on sides of $\mathcal{G}_0$, three of the four surrounding small squares are in $\mathcal{Q}$. At the one inside a big square, only s is. So none of them is bad. Added squares at different bad vertices sit at different corners of a big square, so they do not touch. Hence $\mathcal{Q}$ has no bad vertices. (i)–(iv) persist: the added squares lie in squares of $\mathcal{G}_0$ adjacent to $\mathcal{R}$, within $3\delta < d$ of F_1. The new boundary sides lie in squares of $\mathcal{G}_0$ not in $\mathcal{R}$, so they miss F_1, and also F_2. So $\partial \bigcup Q_j \subset \Omega$.

Step (v). Every boundary vertex of $\bigcup Q_j$ now meets exactly two boundary sides. With the

orientation leaving $\bigcup Q_j$ on the left, one of them enters the vertex and the other leaves it (check the cases of one, two adjacent, or three surrounding squares). Starting from a boundary side $[a_1, a_2]$ we get a unique continuation $[a_2, a_3], [a_3, a_4], \ldots$. There are finitely many sides, so some vertex repeats; let a_m be the first repetition, $a_m = a_k$ with $k < m$. If $k > 1$, then a_k meets the three distinct sides $[a_{k-1}, a_k]$, $[a_k, a_{k+1}]$, $[a_{m-1}, a_m]$, which is impossible. So $k = 1$ and $a_1 \cdots a_m$ is a simple closed polygon. Removing its sides and repeating exhausts the boundary sides, giving finitely many disjoint simple closed polygons (disjoint because each vertex lies on only one of them). ■

Proof of Theorem 1.2, converse. Let Ω be bounded and simply connected. If Ω^c were disconnected, the proof of Theorem 1.4 produces a closed curve $\gamma \subset \Omega$ and $w \notin \Omega$ with $W_\gamma(w) \neq 0$. This contradicts the first part of Theorem 1.4, since Ω is simply connected. ■

2 The Jordan curve theorem

Motivation. Contour integration constantly refers to "the interior of a curve", and the Jordan curve theorem guarantees that such an interior exists for every simple closed curve. We prove it here and deduce a general form of Cauchy's theorem.

The curves in the statements are piecewise-smooth, but the proofs also use curves that are merely continuous (the curves γ_η below).

Theorem 2.1 (Jordan arc theorem)

Let γ be a simple (non-closed) piecewise-smooth curve in the plane. Then γ^c is an open connected set whose boundary is precisely γ.

Theorem 2.2 (Jordan curve theorem)

Let γ be a simple closed piecewise-smooth curve in the plane. Then γ^c consists of two disjoint connected open sets. Exactly one of them is bounded and simply connected; it is the **interior** Ω_γ of γ. The other is unbounded; it is the **exterior** $\mathcal{U}_\gamma$. With the appropriate orientation of γ,

$$W_\gamma(z) = \begin{cases} 1 & z \in \Omega_\gamma, \\ 0 & z \in \mathcal{U}_\gamma. \end{cases}$$

Remark

Both theorems hold for merely continuous curves, but the proofs are harder; the piecewise-smooth case suffices for many applications.

Theorem 2.3

If f is holomorphic in the interior Ω_γ of a simple closed curve γ, then $\int_\eta f(\zeta)\, d\zeta = 0$ for every closed curve $\eta \subset \Omega_\gamma$.

Proof [Supplement]. Ω_γ is a simply connected region by Theorem 2.2, so this is Cauchy's theorem, Corollary 5.3 of Chapter 3. ■

Idea of the proof of Theorem 2.1. γ^c is open, so it suffices to show it is pathwise connected. Join $z, w \notin \gamma$ by a curve. If it meets γ, first go from z to a point z' and from w to a point w' near

γ without meeting γ. Then join z' to w' by travelling "parallel" to γ, going around an end-point if necessary. If γ parametrizes a smooth piece, $\gamma'(t) \neq 0$ is tangent to it, $i\gamma'(t)$ is normal, and $\gamma(t) + i\eta\gamma'(t)$ is a "parallel" curve.

In the next three lemmas and two propositions, γ_0 is a *simple smooth* curve with arc-length parametrization $\gamma : [0, L] \to \mathbb{C}$, $|\gamma'(t)| = 1$ (every smooth curve has one). An *interior point* of γ_0 is a point $\gamma(t)$ with $0 < t < L$ (not to be confused with the interior of Theorem 2.2).

Lemma 2.4

For $\eta \in \mathbb{R}$ let γ_η be the continuous curve $\gamma_\eta(t) = \gamma(t) + i\eta\gamma'(t)$, $0 \le t \le L$. There is $\eta_1 > 0$ such that $\gamma_0 \cap \gamma_\eta = \emptyset$ whenever $0 < |\eta| < \eta_1$.

Proof.

Step Local version. For $s, t \in [0, L]$,

$$\gamma_\eta(t) - \gamma(s) = \int_s^t \gamma'(u)\,du + i\eta\gamma'(t) = \int_s^t [\gamma'(u) - \gamma'(t)]\,du + (t - s + i\eta)\gamma'(t).$$

By uniform continuity of γ' there is $\delta > 0$ with $|\gamma'(x) - \gamma'(y)| < 1/2$ for $|x - y| < \delta$. If $|s - t| < \delta$, then using $|\gamma'(t)| = 1$ and $2|a + ib| \ge |a| + |b|$,

$$|\gamma_\eta(t) - \gamma(s)| \ge |t - s + i\eta| - \frac{|t - s|}{2} \ge \frac{|\eta|}{2}.$$

So $\gamma_\eta(t) \neq \gamma(s)$ when $|t - s| < \delta$ and $\eta \neq 0$.

Step Global version. Choose $0 = t_0 < \cdots < t_n = L$ with $t_{k+1} - t_k < \delta/2$, and set $I_k = \{t : |t - t_k| \le \delta/4\}$, $J_k = \{t : |t - t_k| < \delta/2\}$, $J_k' = \{t \in [0, L] : |t - t_k| \ge \delta/2\}$ (intersected with $[0, L]$). The I_k cover $[0, L]$. If $t \in I_k$ and $s \in J_k$ then $|t - s| < \delta$, so

$$\gamma_\eta(I_k) \cap \gamma(J_k) = \emptyset \qquad (\eta \neq 0). \tag{3}$$

Since γ_0 is simple, the compact sets $\gamma(I_k)$ and $\gamma(J_k')$ are disjoint, at distance $d_k > 0$. For $t \in I_k$, $s \in J_k'$,

$$|\gamma_\eta(t) - \gamma(s)| \ge |\gamma(t) - \gamma(s)| - |\eta| \ge d_k - |\eta| \ge d_k/2 \quad \text{if } |\eta| \le d_k/2,$$

so

$$\gamma_\eta(I_k) \cap \gamma(J_k') = \emptyset \qquad (|\eta| < d_k/2). \tag{4}$$

With $\eta_1 = \min_k d_k/2$, (3) and (4) give $\gamma_0 \cap \gamma_\eta = \emptyset$ for $0 < |\eta| < \eta_1$. ■

Lemma 2.5

Suppose $z \notin \gamma_0$ and $z_0 \in \gamma_0$ is a point closest to z with $z_0 = \gamma(t_0)$, $0 < t_0 < L$ (this holds if z is closer to an interior point of γ_0 than to either end-point). Then $z = \gamma(t_0) + i\eta\gamma'(t_0)$ for some real $\eta \neq 0$, i.e. $z \in \gamma_\eta$.

Proof. Near t_0, $z - \gamma(t) = z - \gamma(t_0) - \gamma'(t_0)(t - t_0) + o(|t - t_0|)$, so

$$|z - z_0|^2 \le |z - \gamma(t)|^2 = |z - z_0|^2 - 2(t - t_0)\,\mathrm{Re}\,[(z - \gamma(t_0))\overline{\gamma'(t_0)}] + o(|t - t_0|).$$

Since $t - t_0$ takes both signs, $\mathrm{Re}[(z - \gamma(t_0))\overline{\gamma'(t_0)}] = 0$, i.e. $(z - \gamma(t_0))\overline{\gamma'(t_0)} = i\eta$ with $\eta \in \mathbb{R}$. As $|\gamma'(t_0)| = 1$, $\overline{\gamma'(t_0)} = 1/\gamma'(t_0)$, so $z - \gamma(t_0) = i\eta\gamma'(t_0)$; and $\eta \neq 0$ because $z \notin \gamma_0$. ■

If $z \in \gamma_\eta$ and $w \in \gamma_\epsilon$ ($\eta, \epsilon \neq 0$) are obtained as in Lemma 2.5, we say z, w are on the **same side** of γ_0 if η, ϵ have the same sign, and on **opposite sides** otherwise. We do not define "the two sides of γ_0"; the notion refers to the chosen representations, and we do not claim the two cases are exclusive. Points on the same side can be joined almost directly by a curve parallel to γ_0. Points on opposite sides require going around an end-point.

Proposition 2.6

Let A, B be the end-points of γ_0 and K a compact set with $\gamma_0 \cap K = \emptyset$ or $\gamma_0 \cap K = \{A, B\}$ (or a subset thereof). If $z, w \notin \gamma_0$ lie on the same side of γ_0, and each is closer to an interior point of γ_0 than to K or to the end-points of γ_0, then z and w can be joined by a continuous curve in the complement of $K \cup \gamma_0$.

Proof. By Lemma 2.5, with $z_0 = \gamma(t_0)$, $w_0 = \gamma(s_0)$ interior points closest to z, w,

$$z = \gamma(t_0) + i\eta_0\gamma'(t_0), \qquad w = \gamma(s_0) + i\epsilon_0\gamma'(s_0),$$

with η_0, ϵ_0 of the same sign, say positive, and say $t_0 \le s_0$.

Step Normal segments. The half-open segment from z to z_0 lies in $(K \cup \gamma_0)^c$. If p is on it with $p \neq z_0$ and $p \in K \cup \gamma_0$, then $|z - p| < |z - z_0| = d(z, K \cup \gamma_0)$, which is absurd. So for all small $\eta > 0$, z can be joined to $z_\eta = \gamma(t_0) + i\eta\gamma'(t_0)$, and likewise w to $w_\eta = \gamma(s_0) + i\eta\gamma'(s_0)$.

Step Parallel curve. Choose $\eta < \eta_1$ (Lemma 2.4) and η smaller than the distance from K to $\{\gamma(t) : t_0 \le t \le s_0\}$. This distance is positive since this arc avoids A, B. Then $\{\gamma_\eta(t) : t_0 \le t \le s_0\}$ joins z_η to w_η. It misses γ_0 by Lemma 2.4, and it misses K since $|\gamma_\eta(t) - \gamma(t)| = \eta$. ∎

Lemma 2.7

There is $\eta_2 > 0$ such that the half-neighborhood

$$N = \{\gamma(L) + \eta e^{i\theta}\gamma'(L) : -\pi/2 \le \theta \le \pi/2,\ 0 < \eta < \eta_2\}$$

is disjoint from γ_0.

Proof. As in Lemma 2.4,

$$\gamma(L) + \eta e^{i\theta}\gamma'(L) - \gamma(t) = \int_t^L [\gamma'(u) - \gamma'(L)]\, du + (L - t + \eta e^{i\theta})\gamma'(L).$$

Choose δ with $|\gamma'(u) - \gamma'(L)| < 1/2$ for $|u - L| < \delta$. Since $\cos\theta \ge 0$, $|L - t + \eta e^{i\theta}|^2 \ge (L-t)^2 + \eta^2 \ge \frac{1}{2}(L - t + \eta)^2$. So for $L - \delta < t \le L$,

$$|\gamma(L) + \eta e^{i\theta}\gamma'(L) - \gamma(t)| \ge \frac{L - t + \eta}{\sqrt{2}} - \frac{L - t}{2} \ge \frac{\eta}{\sqrt{2}} > 0.$$

Thus $\gamma(t) \notin N$ for $L - \delta < t \le L$. Finally, take η_2 less than the distance from $\gamma(L)$ to $\{\gamma(t) : 0 \le t \le L - \delta\}$, which is positive since γ_0 is simple. ∎

Proposition 2.8

Let A be an end-point of γ_0 and K a compact set with $\gamma_0 \cap K = \emptyset$ or $\gamma_0 \cap K = \{A\}$. If $z, w \notin \gamma_0$ are closer to interior points of γ_0 than to K or to the end-points of γ_0, then z and w can be joined by a continuous curve in the complement of $\gamma_0 \cup K$.

Proof. If z, w are on the same side, apply Proposition 2.6. So let them be on opposite sides, and (reversing the parametrization if needed) $A = \gamma(0)$, so the other end-point $B = \gamma(L) \notin K$. Write

$$z = \gamma(t_0) + i\eta_0\gamma'(t_0), \qquad w = \gamma(s_0) - i\epsilon_0\gamma'(s_0), \qquad \eta_0, \epsilon_0 > 0,$$

and let $\tau = \min(t_0, s_0) > 0$. Choose $\eta > 0$ smaller than η_1, η_2 (Lemmas 2.4, 2.7), and the distance from K to $\{\gamma(t) : \tau \le t \le L\}$ (positive since this arc avoids A). As in Proposition 2.6, join z to $\gamma_\eta(t_0)$ by a normal segment. Then follow γ_η from t_0 to L to reach $\gamma(L) + i\eta\gamma'(L)$. Then follow

the half-circle $\gamma(L) + \eta e^{i\theta}\gamma'(L)$, θ from $\pi/2$ down to $-\pi/2$, inside N, to reach $\gamma(L) - i\eta\gamma'(L)$. Then follow $\gamma_{-\eta}$ from L back to s_0, and the normal segment to w. The parallel pieces miss γ_0 by Lemma 2.4; the half-circle misses γ_0 by Lemma 2.7. All pieces lie within η of $\{\gamma(t) : \tau \le t \le L\}$, so they miss K. ■

Proof of Theorem 2.1.

Step Boundary. $O = \gamma^c$ is open and $\partial O \subset \gamma$. Every point of a smooth piece of γ is a limit of points of O: by Lemma 2.4 (applied to a small sub-arc, and noting the rest of γ is at positive distance), $\gamma(t) + i\eta\gamma'(t) \notin \gamma$ for small $\eta \neq 0$. Since ∂O is closed and the smooth points are dense in γ, $\partial O = \gamma$.

Step Connectedness: induction on the number of smooth pieces. First let γ be smooth, with end-points A, B, and let $Z, W \notin \gamma$. Join Z to W by a polygonal path Λ avoiding A and B. If Λ misses γ we are done. Otherwise let p and q be the first and last points of Λ on γ; they are interior points of γ. Points z of Λ just before p satisfy $d(z, \gamma) \le |z - p| \to 0$, while $|z - A|, |z - B|$ stay bounded below. So we may take such a $z \notin \gamma$ closer to an interior point of γ than to A, B, joined to Z along Λ without meeting γ. Similarly take w just after q. Proposition 2.8 with $K = \emptyset$ joins z to w in γ^c.

Now suppose the result holds for curves of $n-1$ smooth pieces, and let $\gamma = K \cup \gamma_0$ with K the union of $n-1$ consecutive pieces and γ_0 smooth. Then K is compact and $K \cap \gamma_0$ is a single end-point A of γ_0. Given $Z, W \notin \gamma$, the induction hypothesis gives a curve from Z to W missing K. Since K^c is open and connected, we may take it polygonal and avoiding the other end-point of γ_0. If it meets γ_0, it does so at interior points. As above, take z (resp. w) just before the first (resp. after the last) meeting point; it is closer to an interior point of γ_0 than to K or to the end-points, since its distance to K is bounded below. Proposition 2.8 joins z to w in $(\gamma_0 \cup K)^c = \gamma^c$. ■

Proof of Theorem 2.2.

Step At most two components. Fix W outside a large disc containing γ, and let $\mathcal{U}$ be the set of points joined to W by a continuous curve in γ^c. $\mathcal{U}$ is open and connected, and contains the exterior of the disc. Let $\Omega = \gamma^c - \mathcal{U}$; we show Ω is connected. Remove a smooth piece γ_0 (with end-points A, B) from γ, and let K be the remaining arc, so $K \cap \gamma_0 = \{A, B\}$. For $Z \in \Omega$, Theorem 2.1 applied to K gives a curve Λ_Z from Z to W missing K. Since $Z \notin \mathcal{U}$, Λ_Z meets γ_0, necessarily at interior points. As in the proof of Theorem 2.1, choose $z, w \in \Lambda_Z$ closer to interior points of γ_0 than to K or to A, B, such that the pieces of Λ_Z from Z to z and from w to W lie in γ^c. Then z, w are on opposite sides of γ_0. Otherwise Proposition 2.6 would join z to w in γ^c, hence Z to W, contradicting $Z \notin \mathcal{U}$. If $Z_1 \in \Omega$ is another point, with corresponding z_1, w_1 on opposite sides, then z and z_1 are on the same side. Otherwise z_1 is on the same side as w, and Proposition 2.6 joins Z_1 to W in γ^c, a contradiction. So Proposition 2.6 joins z to z_1 in γ^c, and Z, Z_1 lie in the same component. Thus γ^c has at most two components, $\mathcal{U}$ and (if non-empty) Ω.

Step Exactly two components. By Lemma 1.3 it suffices to find points with different winding numbers. Parametrize γ and fix $z_0 = \gamma(t_0)$ interior to a smooth piece, $v = \gamma'(t_0) \neq 0$. For $\epsilon > 0$ let

$$z_\epsilon = \gamma(t_0) + i\epsilon v, \qquad w_\epsilon = \gamma(t_0) - i\epsilon v.$$

For small ϵ these avoid γ, and the normal segments $\{\gamma(t_0) \pm isv : 0 < s \le \epsilon\}$ lie in γ^c (Lemma 2.4 for the piece, plus the positive distance from $\gamma(t_0)$ to the rest of γ). By Lemma 1.3(ii),

$$\Delta = |W_\gamma(z_\epsilon) - W_\gamma(w_\epsilon)|$$

is a constant integer for all small $\epsilon > 0$. With $a(t) = \gamma(t) - \gamma(t_0)$,

$$\frac{\gamma'(t)}{\gamma(t) - z_\epsilon} - \frac{\gamma'(t)}{\gamma(t) - w_\epsilon} = \frac{2i\epsilon v\, \gamma'(t)}{a(t)^2 + \epsilon^2 v^2}.$$

Since $a(t) = v(t - t_0) + o(|t - t_0|)$ and $\gamma'(t) = v + o(1)$,

$$\frac{2i\epsilon v\gamma'(t)}{a(t)^2 + \epsilon^2 v^2} = \frac{2i\epsilon}{(t - t_0)^2 + \epsilon^2}\,(1 + \rho(t, \epsilon)),$$

where, given $\sigma > 0$, there is $\delta > 0$ with $|\rho(t,\epsilon)| < \sigma$ for $|t - t_0| \le \delta$, uniformly in ϵ. (Indeed $a^2 + \epsilon^2 v^2 = v^2[(t-t_0)^2 + \epsilon^2] + o((t-t_0)^2)$ and $o((t-t_0)^2)/[(t-t_0)^2 + \epsilon^2] \to 0$.) Split the integral defining $W_\gamma(z_\epsilon) - W_\gamma(w_\epsilon)$ into $|t - t_0| \ge \delta$ and $|t - t_0| < \delta$. On the first part $\gamma(t)$ stays at positive distance from $\gamma(t_0)$, while $z_\epsilon, w_\epsilon \to \gamma(t_0)$, so this part tends to 0 as $\epsilon \to 0$. On the second,

$$\frac{1}{2\pi i}\int_{|t-t_0|<\delta} \frac{2i\epsilon\, dt}{(t-t_0)^2 + \epsilon^2} = \frac{1}{\pi}\int_{-\delta/\epsilon}^{\delta/\epsilon} \frac{ds}{s^2+1} = \frac{2}{\pi}\arctan(\delta/\epsilon) \to 1,$$

and the ρ-term is at most $\frac{\sigma}{\pi}\int_{\mathbb{R}} \frac{ds}{s^2+1} = \sigma$ in absolute value. Letting $\epsilon \to 0$ gives $|\Delta - 1| \le \sigma$ for every $\sigma > 0$, so $\Delta = 1$.

Hence z_ϵ and w_ϵ lie in different components, so γ^c has exactly the two components $\mathcal{U}$ and Ω. $\mathcal{U}$ is unbounded with $W_\gamma = 0$ there (Lemma 1.3(iii)), and Ω is bounded. Since $\Delta = 1$, $W_\gamma = \pm 1$ on Ω, and after possibly reversing orientation $W_\gamma = 1$ there. Every smooth point of γ is a limit of the z_ϵ and of the w_ϵ, hence of points of both components. Smooth points are dense in γ, and the boundary of each component lies in γ. So $\gamma = \partial\Omega = \partial\mathcal{U}$. Write $\Omega_\gamma = \Omega$ and $\mathcal{U}_\gamma = \mathcal{U}$.

Step Ω_γ is simply connected. By Theorem 1.2 it suffices that $\Omega_\gamma^c = \gamma \cup \mathcal{U}_\gamma$ is connected. Suppose $\Omega_\gamma^c = F_1 \cup F_2$ with F_1, F_2 closed, disjoint, non-empty, and let $O_i = \mathcal{U}_\gamma \cap F_i$. The sets O_1, O_2 are disjoint and $\mathcal{U}_\gamma = O_1 \cup O_2$.

O_1 is open. Let $z \in O_1$; small balls B at z lie in $\mathcal{U}_\gamma \subset F_1 \cup F_2$. If every such ball met F_2, then $z \in F_2$ (F_2 closed), contradicting $F_1 \cap F_2 = \emptyset$. So some ball lies in F_1, hence in O_1. Likewise O_2 is open.

$O_1 \neq \emptyset$. Otherwise $F_1 \subset \gamma$ and $\mathcal{U}_\gamma \subset F_2$. Take $z \in F_1 \subset \gamma$. Since $\gamma = \partial\mathcal{U}_\gamma$, every ball at z meets $\mathcal{U}_\gamma \subset F_2$, so $z \in F_2$, a contradiction. Likewise $O_2 \neq \emptyset$.

This disconnects $\mathcal{U}_\gamma$, a contradiction. This completes the proof of the Jordan curve theorem for piecewise-smooth curves. ∎

2.1 Proof of a general form of Cauchy's theorem

Theorem 2.9

If f is holomorphic in an open set containing a simple closed piecewise-smooth curve γ and its interior, then $\int_\gamma f = 0$.

Let $\mathcal{O} \supset \gamma \cup \Omega_\gamma$ be open with f holomorphic in $\mathcal{O}$. The idea is to build a closed curve $\Gamma \subset \Omega_\gamma$ so close to γ that $\int_\gamma f = \int_\Gamma f$. The right side vanishes by Theorem 2.3, since f is holomorphic in the simply connected set Ω_γ. Near smooth parts of γ, Γ is essentially a parallel curve γ_η (Lemma 2.4); near the junctions of smooth parts, Γ follows arcs of circles.

Lemma 2.10

Let $\gamma : [0,1] \to \mathbb{C}$ be a simple smooth curve. For all sufficiently small $\epsilon > 0$ the circle C_ϵ of center $\gamma(0)$ and radius ϵ meets γ in exactly one point.

Proof. Assume $\gamma(0) = 0$. Since $\gamma(1) \neq 0$, the intermediate value theorem applied to $|\gamma(t)|$ gives at least one intersection point for $\epsilon < |\gamma(1)|$. Suppose the claim fails: there are $\epsilon_j \to 0$ such that $|\gamma(t)| = \epsilon_j$ has two distinct solutions. Since $|\gamma(t)| \ge c(\tau) > 0$ for $t \ge \tau$ (simplicity), both solutions tend to 0. By the mean value theorem applied to $h(t) = |\gamma(t)|^2$, there are $t_j \to 0$, $t_j > 0$, with $h'(t_j) = 2\,\mathrm{Re}[\gamma'(t_j)\overline{\gamma(t_j)}] = 0$. But $\gamma(t) = \gamma'(0)t + t\varphi(t)$ and $\gamma'(t) = \gamma'(0) + \psi(t)$ with $\varphi, \psi \to 0$ as $t \to 0$, so

$$\mathrm{Re}[\gamma'(t)\overline{\gamma(t)}] = |\gamma'(0)|^2 t + o(t),$$

which is $\neq 0$ for small $t > 0$ since $\gamma'(0) \neq 0$. This is a contradiction. ∎

Proof of Theorem 2.9. Choose ϵ so small that the open set U of points at distance $< \epsilon$ from γ is contained in $\mathcal{O}$. Let $P_1, \ldots, P_n$ be the consecutive junction points of the smooth parts of γ. Choose $\delta < \epsilon/10$ so small that each circle C_j centered at P_j of radius δ meets γ in exactly two points. This is possible by Lemma 2.10, applied to the two smooth parts meeting at P_j; the rest of γ is at positive distance from P_j. The two points determine two arcs of C_j, exactly one of which, $\mathcal{C}_j$, has its interior in Ω_γ. Indeed, for a smooth part with end-point P_j and small η, the curves γ_η and $\gamma_{-\eta}$ lie on opposite sides of γ, hence in different components of γ^c (proof of Theorem 2.2), and both meet C_j. The disc D_j of center P_j and radius 2δ lies in U, hence in $\mathcal{O}$.

Let B_j be the end-point of $\mathcal{C}_j$ on the smooth part γ_j from P_j to P_{j+1}, and A_{j+1} the end-point of $\mathcal{C}_{j+1}$ on γ_j. Take a chain of discs $\mathcal{D} = \{D_0, \ldots, D_K\}$ in U covering γ, with $D_k \cap D_{k+1} \neq \emptyset$, $D_0 = D_K$, and with the discs D_j among them. By Lemma 2.4 there is a continuous curve Γ_j in Ω_γ and in the union of the discs, running parallel to γ_j, from a point near B_j on $\mathcal{C}_j$ to a point near A_{j+1} on $\mathcal{C}_{j+1}$. Since γ is only C^1, Γ_j need not be smooth; replacing it by a nearby polygonal line we may assume it is. Joining Γ_j to Γ_{j+1} along a piece of $\mathcal{C}_{j+1}$, and so on, gives a closed piecewise-smooth curve $\Gamma \subset \Omega_\gamma$.

f has a primitive on each disc of $\mathcal{D}$ (Theorem 2.1 of Chapter 2). Arguing as in the proof of Theorem 5.1 of Chapter 3, and comparing γ and Γ disc by disc, gives $\int_\gamma f = \int_\Gamma f$: on consecutive discs the primitives differ by constants, and the endpoint contributions telescope. Since $\Gamma \subset \Omega_\gamma$ and Ω_γ is simply connected, $\int_\Gamma f = 0$ by Theorem 2.3. Hence $\int_\gamma f = 0$. ■